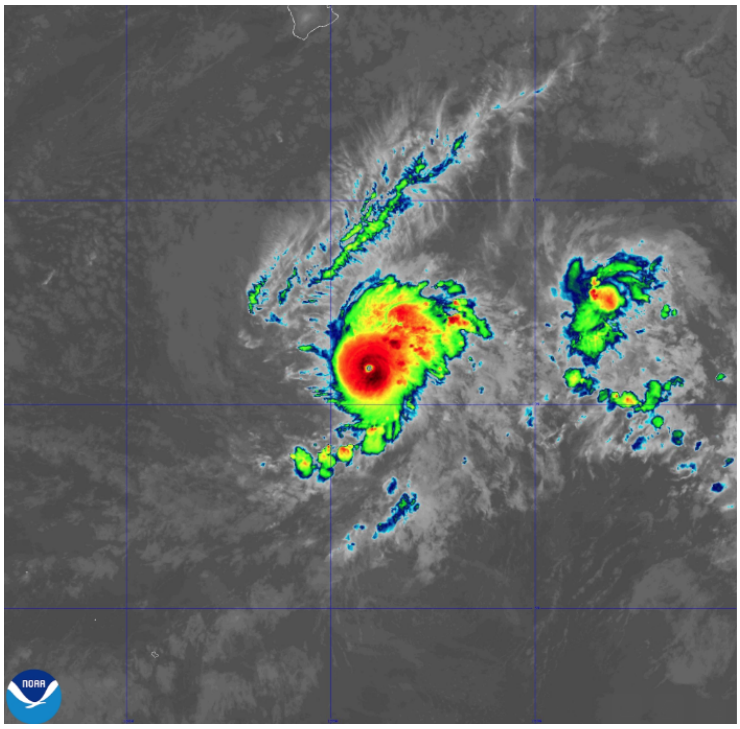
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

HURRICANE IONA

(CP012025)

27 July–1 August 2025

Eric Blake
National Hurricane Center
18 November 2025



GOES-18 INFRARED (BAND 13) IMAGE OF HURRICANE IONA (CENTER) AT 1330 UTC 29 JULY 2025 NEAR PEAK INTENSITY (TROPICAL STORM KELI IS ON THE RIGHT). IMAGE COURTESY NOAA/NESDIS/STAR.

Iona was a relatively long-lived category 4 hurricane (on the Saffir-Simpson Hurricane Wind Scale) for the Central Pacific basin and co-existed with Tropical Storm Keli to its east. Iona dissipated before reaching the International Date Line.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Iona.

Hurricane Iona

27 JULY–1 AUGUST 2025

BEST TRACK

The "best track[2]" positions and intensities for Iona are listed in Table 1. The best track chart is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively.

There were no land-based or ship observations of winds of tropical storm force associated with Iona.

Origin

Iona formed from a fast-moving, low-latitude tropical wave that moved off the coast of west Africa on 10 July, crossing the Windward Islands on 16 July and Central America on 19 July. The wave progressed westward and was part of a breakdown of the Intertropical Convergence Zone well east-southeast of the Hawaiian Islands on 25 July. A low-pressure area formed late that day about the time it crossed into the Central Pacific basin, though convection was limited on 26 July. There was enough organized convection with a well-defined circulation to mark the formation of a tropical depression around 0000 UTC 27 July when it was located about 830 n mi southeast of Hilo.

Peak Intensity and Minimum Pressure

Iona's peak intensity of 115 kt is based on a blend of Dvorak satellite estimates, leaning more on the higher estimates due to the small size of the cyclone. The estimated minimum central pressure of 956 mb is based on the Knaff-Zehr-Courtney pressure-wind relationship.

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[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Central Pacific Hurricane Center (PHFO), the Joint Typhoon Warning Center (JTWC) and the Satellite Analysis Branch (SAB), as well as objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Iona.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Iona.

FORECAST AND WARNING VERIFICATION

The genesis of Iona was adequately forecast, though it lacked some confidence close to the formation time. Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. The system that became Iona was first mentioned in the 7-day TWO a little over 4 days before formation, with a medium (40–60%) chance of formation issued about 2.5 days ahead of genesis. For the 2-day TWO, the system was first given a low (10–30%) chance of formation 42 h prior to genesis and was given a medium chance about 30 h before formation. The system was only given a high (>60%) chance of formation in the 7-day TWO issued at the time of genesis. Figure 4 shows composites of 7-day graphical TWO genesis areas for each category prior to the formation of Iona. The genesis location occurred within 100% of the forecast areas and was also near the center of the outlook areas.

A verification of NHC official track forecasts for Iona is given in Table 3a. Official track forecast errors were lower than the mean official errors for the previous 5-yr period at 12 h and 72 h–120 h, and slightly higher from 24 h–60 h. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. The Google DeepMind (GDMI) ensemble and ECMWF (EMXI) model performed well for Iona, while all of the NCEP regional hurricane models (HMNI, HWFI, HFAI, HFBI) struggled with the track forecast. Note that not all of the GDMI forecasts arrived in time for forecaster use.

A verification of NHC official intensity forecasts for Iona is given in Table 4a. Official intensity forecast errors were much higher than the mean official errors for the previous 5-yr period. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. With the very rapid increase and decrease of Iona's winds, it should be no surprise that the forecasts were less accurate than average. The difficulty of these forecasts can also be seen in the very large OCD5 (climatology/persistence) errors, which from 24 h–60 h were over twice the long-term average errors. Notably, there were six consecutive overlapping 24-h periods where rapid intensification was noted, with a peak of a 60-kt increase in 24 hours. While none of the models were particularly accurate with Iona, they did mostly beat the NHC official forecast. The HFBI and CTCI models had some of the lowest errors for the short-term, while the GDMI was best at longer ranges.

There were no coastal watches or warnings issued for Iona.

ACKNOWLEDGEMENTS

The Central Pacific Hurricane Center and the Joint Typhoon Warning Center provided their fix information for Iona. Dr. Philippe Papin (NHC) created the GTWO verification figure.

Table 1. Best track for Hurricane Iona, 27 July–1 August 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
26 / 0000	10.5	140.9	1009	25	low
26 / 0600	10.7	141.6	1009	25	"
26 / 1200	10.8	142.3	1009	25	"
26 / 1800	10.9	143.1	1009	25	"
27 / 0000	10.9	144.1	1008	25	tropical depression
27 / 0600	10.9	145.1	1007	30	"
27 / 1200	10.9	146.0	1007	30	"
27 / 1800	10.9	146.9	1006	35	tropical storm
28 / 0000	10.9	147.8	1005	40	"
28 / 0600	10.8	148.8	1004	45	"
28 / 1200	10.7	149.7	1001	55	"
28 / 1800	10.7	150.6	996	65	hurricane
29 / 0000	10.8	151.6	988	75	"
29 / 0600	10.8	152.6	970	100	"
29 / 1200	10.8	153.7	956	115	"
29 / 1800	10.9	155.0	960	110	"
30 / 0000	11.0	156.4	968	100	"
30 / 0600	11.2	157.9	976	90	"
30 / 1200	11.4	159.6	991	70	"
30 / 1800	11.6	161.7	1000	55	tropical storm
31 / 0000	11.8	163.8	1002	45	"
31 / 0600	12.1	165.9	1002	45	"
31 / 1200	12.5	167.9	1004	40	"
31 / 1800	13.0	169.8	1004	40	"
01 / 0000	13.4	171.7	1006	35	"
01 / 0600	13.6	173.6	1007	30	tropical depression
01 / 1200	13.8	175.5	1007	30	"
01 / 1800	-	-	-	-	dissipated
29 / 1200	10.8	153.7	956	115	minimum pressure and maximum winds

Table 2. Number of hours in advance of Iona's formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the "Low" category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	42	102
Medium (40%-60%)	30	60
High (>60%)	N/A	0

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Iona, 27 July–1 August 2025. Mean errors for the previous eastern Pacific 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**19.9**	38.1	57.9	73.6	81.6	**78.1**	**82.1**	**104.0**
OCD5	30.1	67.7	122.9	183.7	227.7	287.7	364.0	422.9
Forecasts	20	18	16	14	12	10	6	2
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Iona, 27 July–1 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	19.0	39.2	59.2	77.7	89.6	90.3	91.5	
OCD5	30.3	72.4	128.4	196.3	251.8	332.3	426.4	
GFSI	22.8	43.8	70.4	117.8	164.1	196.8	184.3	
HWFI	34.2	73.1	109.8	153.0	185.2	206.2	211.3	
HMNI	25.7	57.4	93.4	144.6	190.4	237.4	288.5	
HFAI	31.8	72.9	118.2	177.4	225.6	267.0	285.1	
HFBI	27.4	57.8	99.6	153.0	198.5	222.4	228.4	
GDMI	**17.4**	**32.4**	**42.5**	**56.0**	**65.0**	**76.4**	119.4	
EMXI	20.4	**36.7**	**55.7**	**75.7**	**80.2**	**68.9**	**42.5**	
NVGI	22.0	**36.7**	62.7	82.2	102.4	135.8	138.7	
CMCI	23.0	43.6	63.0	**66.4**	**63.9**	**59.3**	**44.9**	
CTCI	30.4	58.7	94.5	135.9	176.3	207.9	247.6	
TVCE	23.8	47.9	76.9	112.9	141.4	165.5	178.4	
TVCX	23.2	47.6	75.4	111.8	139.6	161.5	169.8	
GFEX	19.9	**36.7**	**56.1**	87.8	113.6	127.0	104.9	
TVDG	22.1	44.5	70.3	103.9	130.0	154.1	161.4	
HCCA	21.4	41.4	**58.1**	**77.5**	**79.8**	**82.5**	**72.5**	
FSSE	22.4	41.4	59.9	89.4	105.4	118.6	128.7	
AEMI	21.5	39.6	64.4	96.0	121.3	137.0	122.3	
TABS	19.7	**34.0**	**46.5**	**51.8**	**60.2**	**84.4**	114.9	
TABM	22.5	53.9	96.5	146.3	197.8	253.5	367.6	
TABD	39.0	97.0	167.0	237.7	308.9	384.1	520.7	
Forecasts	18	14	14	12	10	8	4	0

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Iona, 27 July–1 August 2025. Mean errors for the previous eastern Pacific 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	9.0	15.3	25.3	33.6	30.8	21.0	**6.7**	**0.0**
OCD5	13.1	25.0	33.3	43.6	43.3	34.8	19.2	27.0
Forecasts	20	18	16	14	12	10	6	2
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Iona, 27 July–1 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	8.9	17.9	26.1	29.6	25.0	16.2	6.2	
OCD5	13.4	28.9	34.8	40.8	42.3	37.6	23.0	
HWFI	9.7	20.8	30.8	35.1	30.4	19.9	**4.0**	
HMNI	**8.4**	**17.2**	**24.6**	35.7	40.9	33.9	6.2	
HFAI	9.6	18.4	27.1	30.4	28.6	25.0	16.2	
HFBI	**6.8**	**12.1**	**17.4**	**20.2**	**23.9**	20.2	12.2	
GDMI	10.6	18.2	**20.4**	**19.1**	**15.8**	**10.9**	**6.0**	
DSHP	12.8	23.1	**25.9**	**28.1**	**24.2**	18.8	11.5	
LGEM	12.4	22.4	**25.1**	**27.5**	25.3	17.2	8.5	
ICON	10.1	20.1	**25.6**	31.6	30.2	21.9	7.0	
IVCN	**8.5**	**17.4**	**23.5**	**28.5**	27.1	21.5	10.0	
IVDR	**8.2**	**16.9**	**23.3**	**28.2**	27.1	21.2	9.8	
CTCI	**6.9**	**14.9**	**19.6**	**20.4**	**20.3**	17.8	14.8	
GFSI	10.9	21.0	**25.1**	**25.3**	**21.8**	**14.8**	**5.2**	
EMXI	17.4	32.6	38.0	37.3	29.8	**16.0**	**6.0**	
HCCA	9.2	**13.6**	**18.1**	**25.4**	**24.9**	19.5	8.5	
FSSE	**8.2**	**16.9**	**23.7**	**28.4**	25.4	18.0	7.5	
Forecasts	18	14	14	12	10	8	4	0

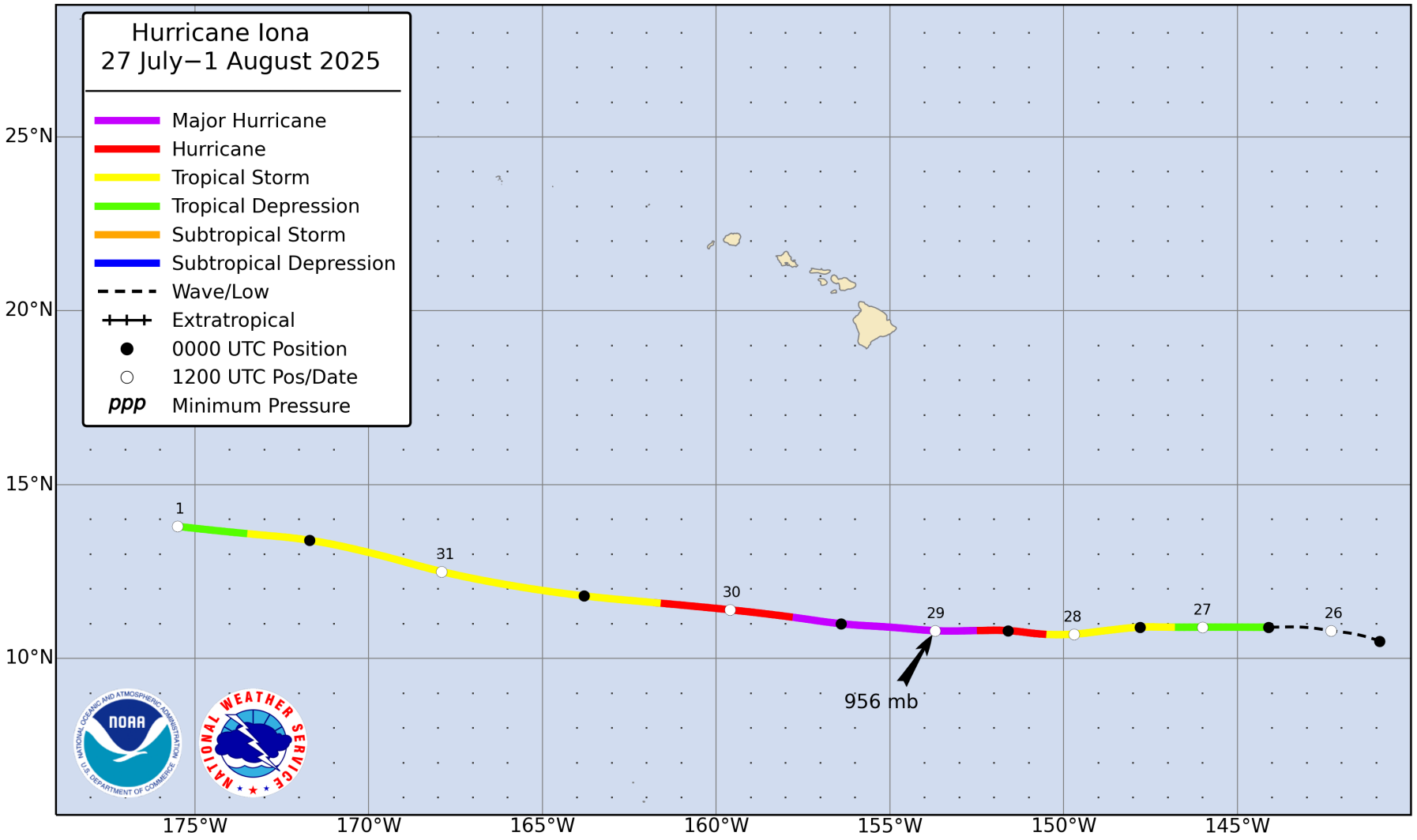


Figure 1. Best track positions for Hurricane Iona, 27 July–1 August 2025.

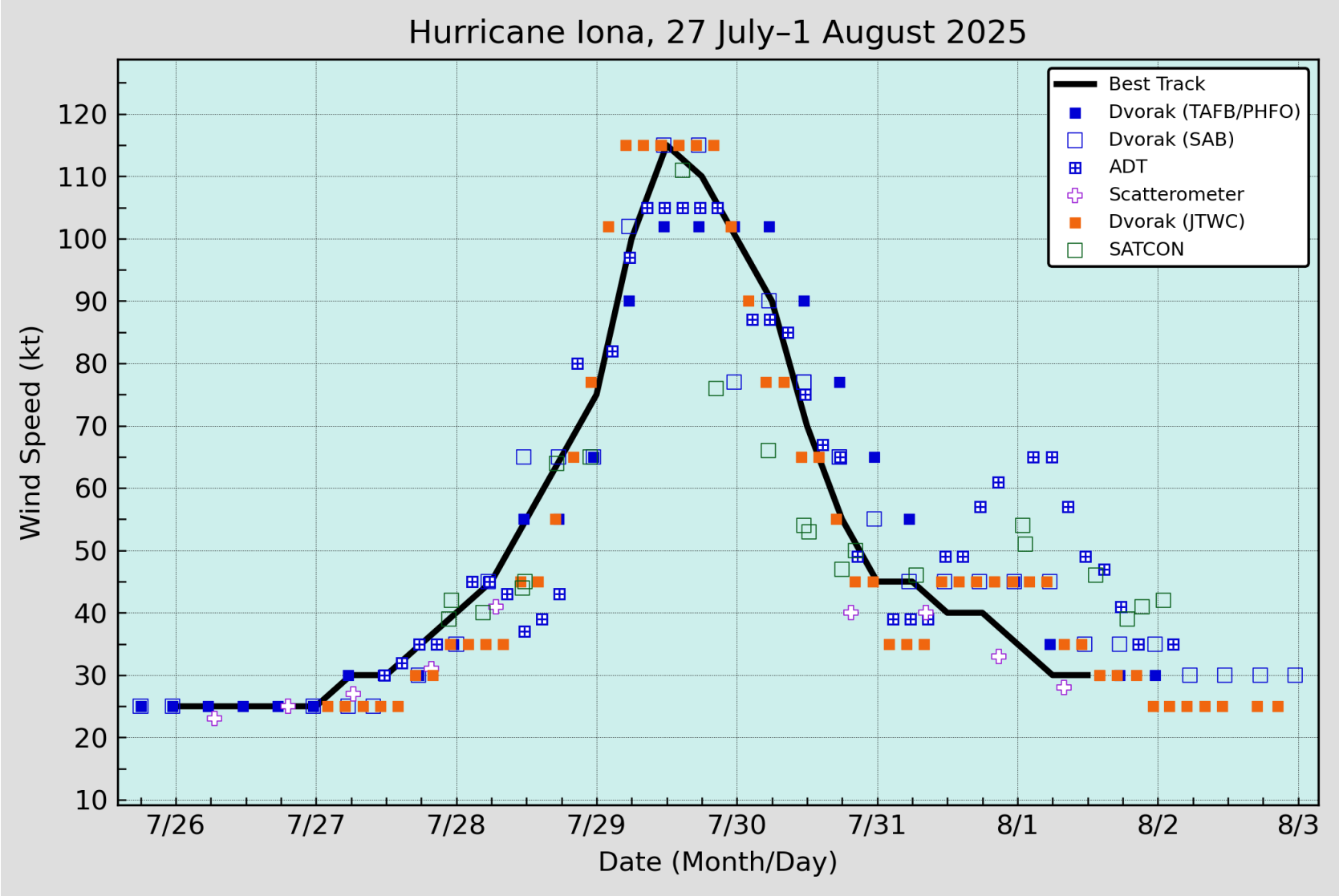


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Iona, 27 July–1 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

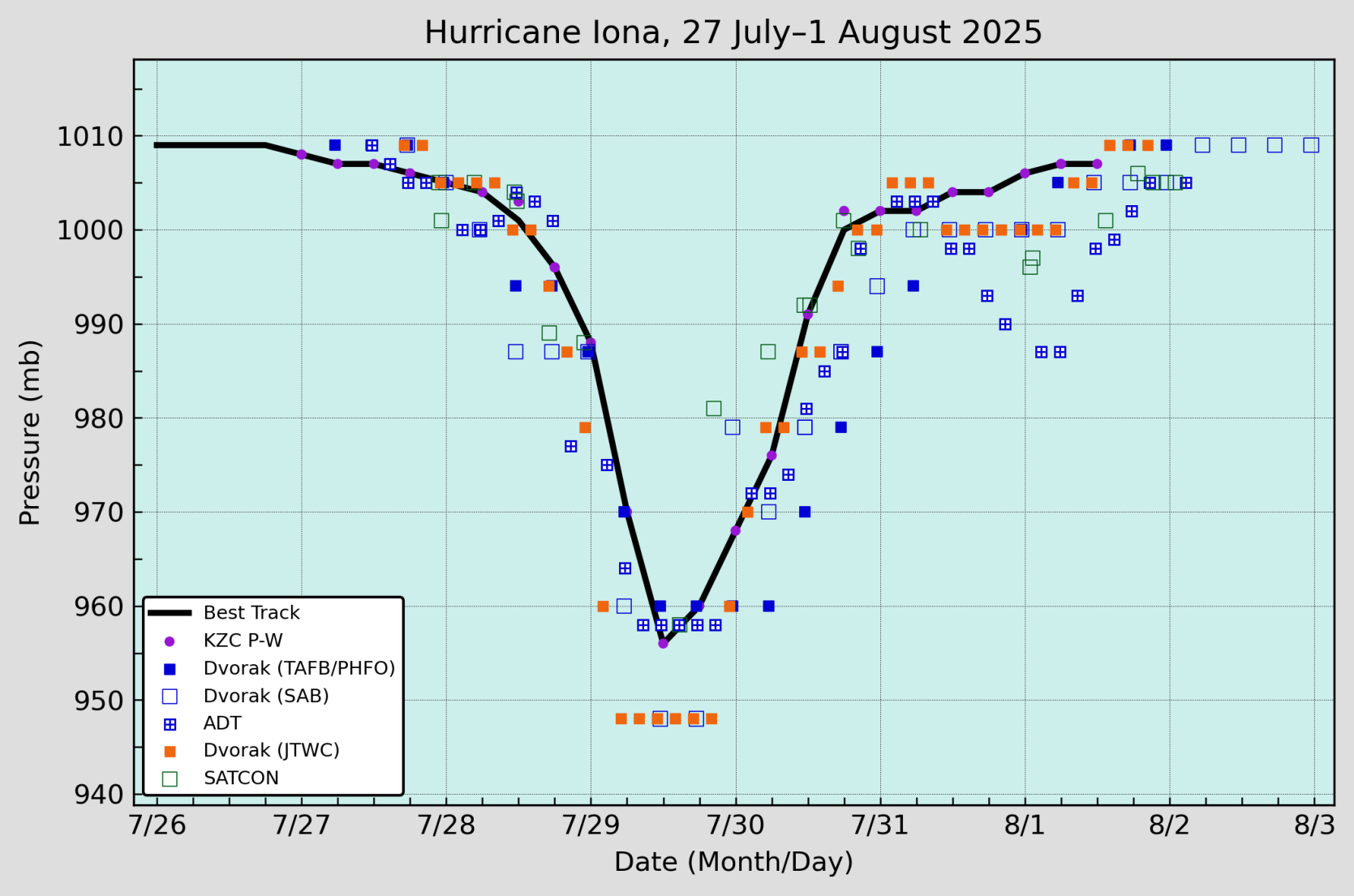


Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Iona, 27 July–1 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

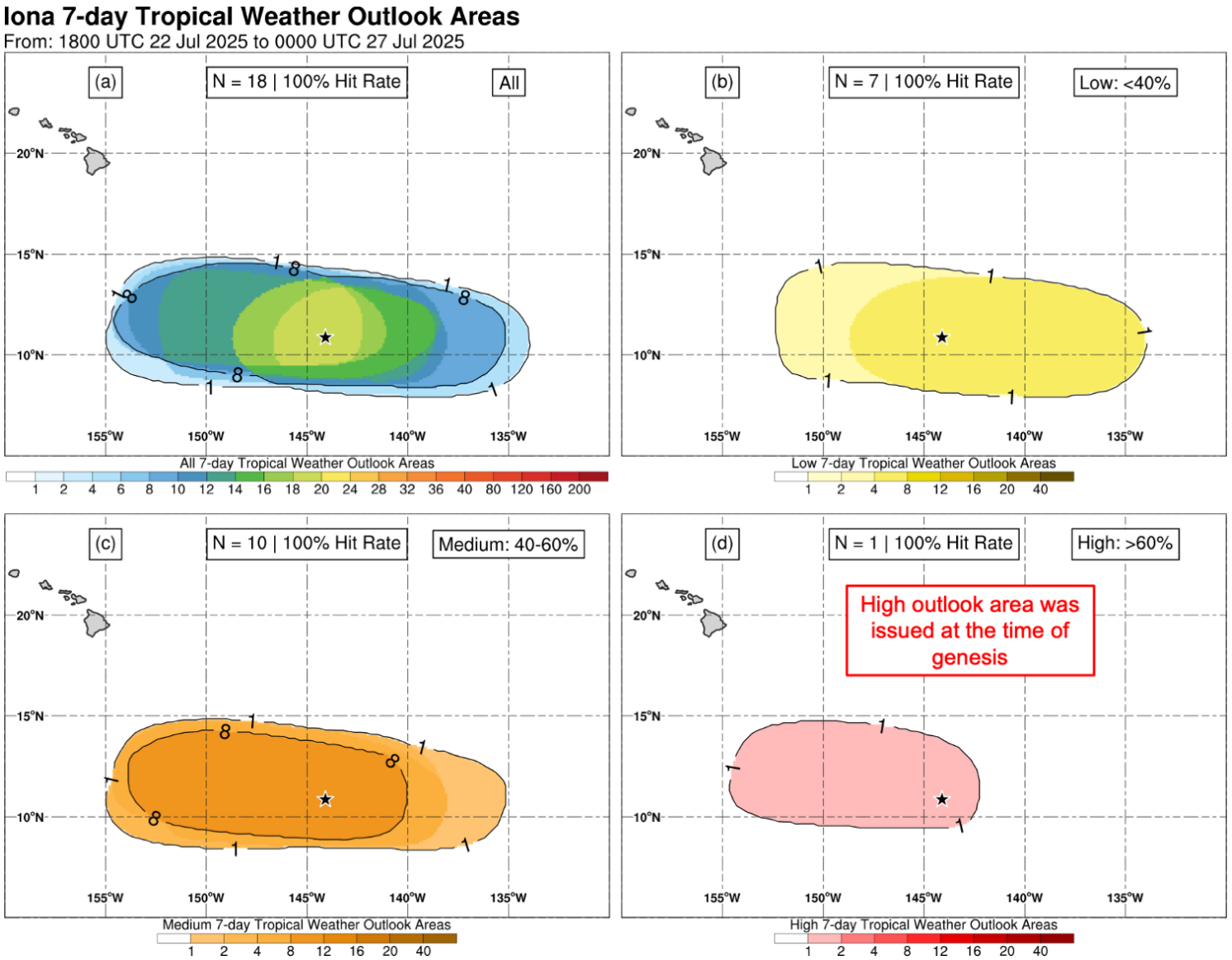


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Hurricane Iona for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.